

Boundary-Localized Genomic Utility and Clinical-Tail Dominance: A Likelihood-Ratio Theory for ZeBRA and Fibrotic Interstitial Lung Disease

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Abstract—Genomic risk and longitudinal clinical history are distinct information channels for forecasting future disease. We study when a static germline genomic channel can add predictive value once a longitudinal clinical-history score has become strongly informative. The central object is the residual genomic likelihood ratio conditional on clinical history. At any fixed clinical stage, the relevant quantity is the high-probability scale of the residual genomic log-likelihood ratio: genomic information can change a likelihood-ratio decision outside the corresponding neighborhood of the clinical boundary only on a low-probability exceptional set. A stronger uniform-tightness condition is required only for a sequential tail-dominance statement as clinical history lengthens. Under a still stronger deterministic boundedness assumption, the boundary band becomes exact, and sufficiently extreme clinical likelihood-ratio thresholds dominate any genomic-only rule whose marginal likelihood ratio is bounded. We evaluate these predictions in a Colorado fibrotic interstitial lung disease (ILD) analysis using ZeBRA, a longitudinal EHR-derived score, and a curated genomic panel containing *MUC5B* rs35705950. Across repeated held-out splits, ZeBRA dominates global discrimination, broad genomic augmentation provides essentially no global FILD/FILA gain after conditioning on ZeBRA, and ZeBRA predicts *MUC5B* carrier status only at chance level. Nevertheless, *MUC5B* shows localized conditional predictive association in restricted ZeBRA operating bands, and across the evaluated two-threshold rescue grid the largest mean held-out sensitivity gain is approximately 4.5 percentage points at essentially matched operational false-positive rate. A new direct empirical likelihood-ratio analysis further shows control-side high-probability residual-genomic bounds of about 0.87 and 0.91 log-LR units at the 95% and 99% levels, with clinical log-likelihood clearly dominating residual genomic scale in the extreme ZeBRA tail. The empirical pattern is therefore not global genomic redundancy but regime-dependent complementarity localized near the clinical decision boundary.

Index Terms—ZeBRA, fibrotic interstitial lung disease, idiopathic pulmonary fibrosis, *MUC5B*, genomics, likelihood ratio, Neyman–Pearson lemma, Bayesian inference, decision boundary.

I. MOTIVATING PROBLEM: ZEBRA AND GENOMIC RISK IN FIBROTIC ILD

Fibrotic ILDs, including idiopathic pulmonary fibrosis (IPF), have a strong but non-Mendelian genetic component. The *MUC5B* promoter variant rs35705950 is among the strongest common genetic risk factors for pulmonary fibrosis [2]–[4]. In the original report, the minor allele was present much more frequently among subjects with IPF than among controls [2]. At the same time, disease emergence is accompanied by a growing

sequence of clinically observable diagnoses, prescriptions, procedures, and comorbidities. ZeBRA is designed to summarize such routinely observed longitudinal clinical histories into a future-risk score without requiring laboratory tests, imaging, or questionnaires [1].

Our recent fibrotic ILD analyses ask whether genomic information improves prediction once ZeBRA is known. In 30 repeated held-out splits of the current operational FILD/FILA endpoint, mean AUC is approximately 0.814 for ZeBRA, 0.566 for the selected genomic panel, and 0.799 for a nonlinear combined model. Because that comparison alone does not isolate incremental information, we also performed a paired regularized analysis in which ZeBRA-only and ZeBRA-plus-genomics models were evaluated on the same held-out subjects. For FILD/FILA, the mean paired AUC increment from the complete genomic panel is only +0.00043, the median increment is 0, and the split-wise 2.5th–97.5th percentile range spans approximately −0.0051 to +0.0041; average precision decreases and both Brier score and log loss worsen after broad genomic augmentation.

The absence of global gain is not explained by ZeBRA reconstructing the major genomic signal. Among subjects with called rs35705950 genotype, ZeBRA predicts *MUC5B* carrier status at chance level (AUC approximately 0.509). A modest pooled enrichment of carriers in the extreme ZeBRA tail disappears after stratification by FILD/FILA status, showing that the pooled association is consistent with both channels tracking disease rather than with clinical history encoding genotype. Yet residual *MUC5B* disease information is strongly heterogeneous across ZeBRA score space. In the ZeBRA band corresponding approximately to 5–10% control FPR, adding *MUC5B* after ZeBRA yields a substantially larger local likelihood-ratio statistic than the converse addition of ZeBRA after *MUC5B*. Across the evaluated two-threshold grid, a rule that consults *MUC5B* only between a stringent upper and lower ZeBRA threshold achieves a largest mean held-out sensitivity gain of approximately 4.5 percentage points while preserving essentially the same realized operational FPR. These observations motivate and correspond to the theory below; they are not taken as proofs of its regularity assumptions.

The apparent paradox is therefore:

A strong genomic risk factor can be biologically important, and can improve decisions for selected

individuals, while contributing almost nothing to global discrimination once a strong clinical-history predictor is available.

We show that this behavior follows naturally from a Bayesian likelihood-ratio formulation under the stochastic residual-evidence and clinical-tail conditions developed below.

II. LATENT-STATE AND INFORMATION-CHANNEL FORMULATION

Let B_t denote the latent biological state at time t , G the germline genomic state, and $E_{0:t}$ the accumulated non-genetic exposure history. The biological state evolves under the influence of G and $E_{0:t}$. For a fixed future horizon $h > 0$, define the disease-risk functional

$$\rho_h(B_t) = \mathbb{P}(T \leq t+h \mid T > t, B_t), \quad (1)$$

where T is the disease-event time. The clinical event sequence $C_{0:t}$ is a noisy observation of the latent biological trajectory $B_{0:t}$. In the idealized Bayesian limit, a clinical-history predictor satisfies

$$Z_h(C_{0:t}) = \mathbb{P}(T \leq t+h \mid C_{0:t}) = \int \rho_h(b) d\mathbb{P}(B_t = b \mid C_{0:t}). \quad (2)$$

Thus ZeBRA need not reconstruct B_t ; it estimates the disease-relevant functional of B_t from the clinical observation channel.

For the theorem it is enough to reduce the problem to binary prediction. Let

$$Y = \mathbf{1}\{T \leq t+h\} \in \{0, 1\}, \quad (3)$$

and write C for the available clinical history. The genomic channel G and clinical channel C are then placed on equal statistical footing through their disease-versus-control likelihood ratios.

III. LIKELIHOOD-RATIO REPRESENTATION

Let $\mathbb{P}_1(\cdot) = \mathbb{P}(\cdot \mid Y = 1)$ and $\mathbb{P}_0(\cdot) = \mathbb{P}(\cdot \mid Y = 0)$. Assume throughout that the case distributions used below are absolutely continuous with respect to their control counterparts, so the relevant likelihood ratios exist as Radon–Nikodym derivatives and satisfy $\mathbb{E}_0[\Lambda] = 1$. Where the deterministic two-sided bounded-residual corollary is invoked, the corresponding conditional genomic laws are mutually absolutely continuous. For readability we write these derivatives using densities with respect to common dominating measures. Define the clinical likelihood ratio

$$\Lambda_C(c) = \frac{p(c \mid Y = 1)}{p(c \mid Y = 0)}, \quad (4)$$

and the residual genomic likelihood ratio conditional on clinical history

$$\Lambda_{G|C}(g; c) = \frac{p(g \mid Y = 1, C = c)}{p(g \mid Y = 0, C = c)}. \quad (5)$$

The full joint likelihood ratio is

$$\Lambda_{C,G}(c, g) = \frac{p(c, g \mid Y = 1)}{p(c, g \mid Y = 0)}. \quad (6)$$

By the probability chain rule,

$$\Lambda_{C,G}(c, g) = \Lambda_C(c) \Lambda_{G|C}(g; c). \quad (7)$$

No conditional-independence assumption is required.

If Z is a calibrated clinical posterior, $Z = \mathbb{P}(Y = 1 \mid C)$, and $\pi = \mathbb{P}(Y = 1)$, Bayes' rule yields

$$\frac{Z}{1-Z} = \frac{\pi}{1-\pi} \Lambda_C(C), \quad (8)$$

so thresholding calibrated ZeBRA is equivalent to thresholding Λ_C . More generally, the same results apply whenever the clinical score is a strictly monotone transformation of Λ_C .

IV. MAIN THEOREM

The boundary-localization result does not require a likelihood-ratio bound that is uniform over all possible clinical-history lengths. At any fixed observed history length, the relevant quantity is simply the high-probability scale of the residual genomic log-likelihood ratio. A stronger uniform condition is needed only for a sequential statement whose bound remains stable as clinical history accumulates.

Notation.: For sequential clinical histories $C_n = (X_1, \dots, X_n)$, write

$$R_{G,n} = \log \Lambda_{G|C_n}(G; C_n) \quad (9)$$

for the residual genomic log-likelihood ratio at stage n . For a clinical likelihood-ratio threshold $\lambda > 0$, define

$$D_{C_n} = \mathbf{1}\{\Lambda_{C_n} \geq \lambda\}, \quad D_{C_n,G} = \mathbf{1}\{\Lambda_{C_n} \Lambda_{G|C_n} \geq \lambda\}. \quad (10)$$

For each fixed n and tolerance $\delta \in (0, 1)$, define the finite-stage residual-evidence scale

$$b_{\delta,n} = \inf \{b \geq 0 : \mathbb{P}_0(|R_{G,n}| > b) \leq \delta\}, \quad (11)$$

whenever this quantile is finite. Thus $b_{\delta,n}$ is a property of the residual genomic evidence distribution at the clinical stage actually being analyzed; no uniformity over n is required for the boundary-localization statement below.

A1-U (Optional uniform tightness of residual genomic evidence). For the stronger sequential tail statement, assume that for each tolerance $\delta \in (0, 1)$ there exists a finite b_δ , independent of n , such that

$$\sup_n \mathbb{P}_0(|R_{G,n}| > b_\delta) \leq \delta. \quad (12)$$

This is stronger than finite-stage boundedness. It allows rare large residual genomic effects but excludes a residual genomic scale that grows without bound as the clinical record lengthens. The empirical analysis below estimates a fixed-score analogue of $b_{\delta,n}$ for *MUC5B*; it does not establish A1-U for arbitrary genomic features or for all n .

Notation	Meaning
$b_{\delta,n}$	Finite-stage $(1 - \delta)$ residual-genomic log-LR scale at a particular clinical-history stage n ; no uniformity over history length is assumed.
b_δ	A single n -independent residual-genomic bound available only under the stronger uniform-tightness condition A1-U.

We use the subscript n precisely to distinguish the empirically relevant finite-stage quantity from the stronger sequential bound.

A2 (Extended clinical-evidence tail). Clinical history is assumed to admit arbitrarily strong disease evidence on rare trajectories. Formally, for arbitrarily large $K > 0$ there exists an n such that

$$\mathbb{P}_0(\log \Lambda_{C_n}(C_n) \geq K) > 0. \quad (13)$$

A2 is an explicit structural assumption, not a consequence of high AUC or of sequential observation alone. The empirical analysis can demonstrate a strong observed tail over the available range, but cannot prove mathematical unboundedness.

Theorem 1 (Finite-stage boundary localization and sequential clinical-tail dominance). *Assume the relevant likelihood ratios are well defined and $b_{\delta,n} < \infty$ at the finite stage under consideration.*

(i) Exact factorization. *For every n ,*

$$\Lambda_{C_n,G}(c_n, g) = \Lambda_{C_n}(c_n) \Lambda_{G|C_n}(g; c_n). \quad (14)$$

(ii) Finite-stage high-probability boundary localization. *For every n , every decision threshold $\lambda > 0$, and every $\delta \in (0, 1)$,*

$$\mathbb{P}_0(D_{C_n,G} \neq D_{C_n}, |\log \Lambda_{C_n} - \log \lambda| > b_{\delta,n}) \leq \delta. \quad (15)$$

Thus, at a fixed clinical stage, genomic information can reverse the clinical decision outside a log-LR neighborhood of half-width $b_{\delta,n}$ only on a control-side exceptional event of probability at most δ .

(iii) Sequential high-probability clinical-tail dominance. *If A1-U also holds, then for every n , every positive upper-tail threshold $K > 0$, and every $\delta \in (0, 1)$,*

$$\mathbb{P}_0\left(\log \Lambda_{C_n} \geq K, \left|\frac{\log \Lambda_{C_n,G} - \log \Lambda_{C_n}}{\log \Lambda_{C_n}}\right| > \frac{b_{\delta}}{K}\right) \leq \delta. \quad (16)$$

The denominator is therefore strictly positive on the event to which the bound is applied. If A2 also holds, positive-probability upper clinical-tail events exist for arbitrarily large K , while $b_{\delta}/K \rightarrow 0$. Apart from the low-probability residual-genomic exception, the relative contribution of residual genomic log-evidence is then arbitrarily small on sufficiently strong disease-evidence tails.

Proof. Part (i) is the chain rule:

$$\Lambda_{C_n,G}(c_n, g) = \frac{p(c_n, g | Y = 1)}{p(c_n, g | Y = 0)} \quad (17)$$

$$= \frac{p(c_n | Y = 1)p(g | Y = 1, c_n)}{p(c_n | Y = 0)p(g | Y = 0, c_n)} \quad (18)$$

$$= \Lambda_{C_n}(c_n) \Lambda_{G|C_n}(g; c_n). \quad (19)$$

For part (ii), if both $|\log \Lambda_{C_n} - \log \lambda| > b_{\delta,n}$ and $|R_{G,n}| \leq b_{\delta,n}$ hold, then adding $R_{G,n}$ cannot change the sign of $\log \Lambda_{C_n} - \log \lambda$. If $\log \Lambda_{C_n} - \log \lambda > b_{\delta,n}$, then

$$\log \Lambda_{C_n,G} - \log \lambda = \log \Lambda_{C_n} - \log \lambda + R_{G,n} > 0, \quad (20)$$

so both rules classify positive. The negative-side case is symmetric. Hence

$$\begin{aligned} \{D_{C_n,G} \neq D_{C_n}, |\log \Lambda_{C_n} - \log \lambda| > b_{\delta,n}\} \\ \subseteq \{|R_{G,n}| > b_{\delta,n}\}, \end{aligned} \quad (21)$$

whose probability is at most δ by the definition of $b_{\delta,n}$.

For part (iii), consider explicitly the upper disease-evidence event $\{\log \Lambda_{C_n} \geq K\}$ with $K > 0$. On

$$\{|R_{G,n}| \leq b_{\delta}\} \cap \{\log \Lambda_{C_n} \geq K > 0\}, \quad (22)$$

we have

$$\left|\frac{\log \Lambda_{C_n,G} - \log \Lambda_{C_n}}{\log \Lambda_{C_n}}\right| = \frac{|R_{G,n}|}{\log \Lambda_{C_n}} \leq \frac{b_{\delta}}{K}. \quad (23)$$

Therefore violation of the relative bound on this upper tail implies $|R_{G,n}| > b_{\delta}$, whose $\mathbb{P}_0$ probability is at most δ uniformly in n by A1-U. A2 ensures only that such positive-probability upper-tail events exist for arbitrarily large K ; it is not derived from the preceding algebra. $\square$

Corollary 1 (Deterministic bounded-residual special case). *Fix a clinical representation C . If there exist constants $0 < m \leq 1 \leq M < \infty$ such that*

$$0 < m \leq \Lambda_{G|C}(g; c) \leq M < \infty \quad (24)$$

for $\mathbb{P}_0$ -almost every relevant (g, c) , then genomic information can change which side of threshold λ an observation occupies only if

$$\frac{\lambda}{M} \leq \Lambda_C(c) \leq \frac{\lambda}{m}. \quad (25)$$

Equivalently,

$$-\log M \leq \log \Lambda_C(c) - \log \lambda \leq -\log m. \quad (26)$$

The deterministic decision-relevant band has fixed width $\log(M/m)$ on the clinical log-likelihood scale. If the same m and M apply uniformly to every C_n , then A1-U also holds with $b_{\delta} = \max\{\log M, -\log m\}$.

Proof. Under (24), $R_G = \log \Lambda_{G|C}$ lies in $[\log m, \log M]$ almost surely. If $\Lambda_C < \lambda/M$, even the largest genomic factor cannot cross the threshold; if $\Lambda_C > \lambda/m$, even the smallest genomic factor cannot move the observation below threshold. This gives (25)–(26). $\square$

Corollary 2 (Posterior-risk form for a calibrated clinical score). *If a clinical score Z is calibrated so that $Z = \mathbb{P}(Y = 1 | C)$, let $\tau \in (0, 1)$ be any decision threshold and define posterior-odds threshold $\kappa = \tau/(1 - \tau)$. Under Corollary 1, genomic information can change the decision only if*

$$\frac{\kappa}{M} \leq \frac{Z}{1 - Z} \leq \frac{\kappa}{m}. \quad (27)$$

Equivalently,

$$-\log M \leq \text{logit}(Z) - \text{logit}(\tau) \leq -\log m. \quad (28)$$

This corollary is a theoretical mapping for a calibrated posterior score. The raw ZeBRA output in the Colorado

cohort is not calibrated as such and is therefore not inserted numerically into this formula.

Proof. Equation (8) shows that posterior odds differ from Λ_C only by the positive constant prior odds $\pi/(1-\pi)$. Substituting into (25) yields the result. $\square$

Proposition 1 (Genomic-only power bound and extreme-specificity clinical superiority). *Assume the marginal genomic likelihood ratio*

$$\Lambda_G(g) = \frac{p(g | Y = 1)}{p(g | Y = 0)} \quad (29)$$

is bounded above by $M_G < \infty$ and A2 holds. Then:

(i) for any genomic-only test $\phi(G) \in [0, 1]$ with false-positive rate $\alpha = \mathbb{E}_0[\phi(G)]$,

$$\boxed{\mathbb{E}_1[\phi(G)] \leq M_G \alpha;} \quad (30)$$

(ii) for any $K > M_G$ with $\alpha_K = \mathbb{P}_0(\Lambda_C \geq K) > 0$, the clinical likelihood-ratio rule

$$\phi_K(C) = \mathbf{1}\{\Lambda_C(C) \geq K\} \quad (31)$$

has

$$\frac{\mathbb{P}_1(\Lambda_C \geq K)}{\mathbb{P}_0(\Lambda_C \geq K)} \geq K > M_G, \quad (32)$$

so at the same FPR α_K its power exceeds that of every genomic-only rule. Moreover $\alpha_K \leq 1/K \rightarrow 0$ as $K \rightarrow \infty$.

Proof. For part (i), using the Radon–Nikodym identity under $\mathbb{P}_0$,

$$\mathbb{E}_1[\phi(G)] = \mathbb{E}_0[\Lambda_G(G)\phi(G)] \quad (33)$$

$$\leq M_G \mathbb{E}_0[\phi(G)] \quad (34)$$

$$= M_G \alpha. \quad (35)$$

For part (ii), define $A_K = \{c : \Lambda_C(c) \geq K\}$. Then

$$\mathbb{P}_1(A_K) = \mathbb{E}_0[\Lambda_C(C)\mathbf{1}\{A_K\}] \quad (36)$$

$$\geq K\mathbb{P}_0(A_K) = K\alpha_K, \quad (37)$$

which gives (32). If $K > M_G$, part (i) implies that any genomic-only test with false-positive rate α_K has power at most $M_G\alpha_K < K\alpha_K \leq \mathbb{P}_1(A_K)$. Markov's inequality applied to $\mathbb{E}_0[\Lambda_C] = 1$ yields $\alpha_K \leq 1/K$. $\square$

Corollary 3 (Genomic post-processing inherits the marginal bound). *Let S be generated from the measured genomic state G by a class-independent Markov kernel $Q(ds | g)$, so that $Y \rightarrow G \rightarrow S$ forms a Markov chain. This includes deterministic scores $S = f(G)$ and randomized post-processing whose additional randomness is independent of disease class conditional on G . If $\Lambda_G \leq M_G$ almost surely under $\mathbb{P}_0$, then $\Lambda_S \leq M_G$ almost surely. Thus no such post-processing of the same genomic information can evade Proposition 1.*

Proof. Under the class-independent channel Q , the score likelihood ratio satisfies

$$\Lambda_S(s) = \mathbb{E}_0[\Lambda_G(G) | S = s]. \quad (38)$$

A conditional expectation of a random variable bounded above by M_G is itself bounded above by M_G . For a stochastic learning

algorithm, this statement is interpreted conditional on the fitted model (or training sample); the deployment-time map from a new subject's G to S must satisfy the stated class-independence condition. $\square$

Proposition 2 (Local decision gain can coexist with negligible global AUC change). *Let $S(C)$ be a clinical score and $S'(C, G)$ an augmented score. Suppose $S' = S$ except on a measurable region B of the clinical–genomic state space. Then*

$$|\text{AUC}(S') - \text{AUC}(S)| \leq \mathbb{P}_1(B) + \mathbb{P}_0(B) - \mathbb{P}_1(B)\mathbb{P}_0(B) \quad (39)$$

$$\leq \mathbb{P}_1(B) + \mathbb{P}_0(B). \quad (40)$$

Hence an augmented rule can produce a substantial change in decisions within a small boundary region while producing an arbitrarily small change in global AUC as the case and control mass of that region tends to zero.

Proof. AUC is the probability that a randomly drawn case receives a higher score than a randomly drawn control, with the usual half-credit convention for ties. If neither member of a case–control pair lies in B , then $S' = S$ for both members and the pairwise ranking contribution is unchanged. Therefore the AUC contribution can differ only when at least one member of the pair lies in B . The probability of that event is

$$1 - (1 - \mathbb{P}_1(B))(1 - \mathbb{P}_0(B)) = \mathbb{P}_1(B) + \mathbb{P}_0(B) - \mathbb{P}_1(B)\mathbb{P}_0(B), \quad (41)$$

which bounds the maximum possible AUC change. $\square$

V. WHY THE CLINICAL CHANNEL CAN HAVE AN EXTENDED TAIL

The theorem does not assume that every later clinical event is informative, nor that disease becomes perfectly observable. It requires only that the clinical channel can accumulate sufficiently strong evidence on some disease trajectories. For a sequential history $C_n = (X_1, \dots, X_n)$, the clinical likelihood ratio factorizes exactly as

$$\Lambda_C(C_n) = \prod_{k=1}^n \frac{p(X_k | Y = 1, X_{<k})}{p(X_k | Y = 0, X_{<k})}, \quad (42)$$

so that

$$\log \Lambda_C(C_n) = \sum_{k=1}^n \log \frac{p(X_k | Y = 1, X_{<k})}{p(X_k | Y = 0, X_{<k})}. \quad (43)$$

This is the standard likelihood-ratio accumulation underlying sequential testing and information accumulation [6], [7]. The genomic state does not acquire new germline variants as disease approaches, whereas the clinical history can acquire additional diagnoses, procedures, prescriptions, and comorbidities. The difference is therefore not that clinical and genomic predictors obey different probability laws; both enter Bayes' rule as evidence. At any fixed clinical stage, Theorem 1(ii) measures the residual genomic scale by a control-side quantile $b_{\delta,n}$. The stronger A1-U condition is needed only to keep that residual scale uniformly tight as the clinical record lengthens; under

Corollary 1 the same idea sharpens to a literal deterministic bound.

Equation (42) also clarifies the sense in which clinical odds may rise sharply: likelihood evidence adds in log space but multiplies in odds space. No claim of exponential growth follows from AUC alone. Nor does the sequential factorization prove A2. It shows only how evidence can accumulate; A2 separately assumes that arbitrarily strong positive clinical-evidence tails are attainable on some trajectories.

VI. INTERPRETATION OF THE THEORETICAL RESULT

The revised theory makes five distinct points that should not be conflated.

A. Global discrimination is not the quantity controlled by the theorem

Theorem 1(ii) controls where genomics can change a threshold decision, not the global ranking of all case-control pairs. If genomic information matters only in a narrow decision band, then only a restricted fraction of pairs can be reordered. Proposition 2 makes the consequence explicit: meaningful local operating-point improvement can coexist with negligible global AUC change.

B. Weak ZeBRA-genotype association does not imply zero conditional genomic disease information

The relevant residual quantity is $\Lambda_{G|C}$, not the marginal dependence between ZeBRA and genotype. ZeBRA can be nearly uninformative about rs35705950 carrier status while *MUC5B* still changes disease odds conditional on a given clinical state.

C. Finite-stage localization does not require uniformity over history length

At a fixed observed clinical stage, Theorem 1(ii) is expressed through the finite-stage residual-evidence quantile $b_{\delta,n}$ and requires no common bound over all n . A1-U is an optional stronger assumption used only for sequential tail dominance. This distinction is important biologically and empirically: the Colorado analysis estimates a score-level residual-genomic scale for *MUC5B*, but does not establish uniform tightness for arbitrary genomic features as clinical history accumulates. Corollary 1 recovers the exact boundary band as a stronger special case.

D. Operating-threshold dependence is a feature, not a defect

Corollary 2 shows, for a calibrated posterior score, that the decision-relevant genomic region is defined relative to the chosen threshold. There is no threshold-free universal “useful band.” Changing the tolerated false-positive rate moves the clinical boundary and therefore changes which patients are potentially reclassified by genomics. Because the raw Colorado ZeBRA score is not calibrated as a posterior probability, this corollary is not applied numerically in the empirical section.

E. Clinical-tail dominance is a stringent-operating-point statement

Proposition 1 and Theorem 1(iii) concern the upper disease-evidence likelihood-ratio tail, explicitly $\{\log \Lambda_{C_n} \geq K > 0\}$. The empirically relevant quantities are therefore direct likelihood-ratio estimates, decision-flip localization, and stringent-specificity LR+ behavior rather than global AUC alone. The data can support the observed tail geometry over the measured range; they do not prove the mathematical unboundedness assumption A2.

VII. EMPIRICAL CORRESPONDENCE IN THE COLORADO ILD ANALYSIS

A. Cohort, provenance, and analysis hierarchy

The empirical study uses a Colorado genomic-clinical cohort with linked longitudinal ZeBRA scores and germline genotype features. The primary operational endpoint is future fibrotic ILD or fibrotic interstitial lung abnormality (FILD/FILA); nonfibrotic ILA and nonfibrotic ILD are retained as ancillary endpoints. Subjects without a usable 104-week ZeBRA score are excluded. In the historical endpoint construction, missing adjudication values are included in the target-zero comparison group, so this group is an operational negative set rather than a fully adjudicated control cohort. False-positive rates are therefore described as operational unless explicitly restricted to adjudicated negatives.

The common repeated-split cohort contains 12,825 subjects. The FILD/FILA endpoint has 254 positives, whereas the nonfibrotic ILA and nonfibrotic ILD endpoints have only 74 and 27 positives, respectively. We therefore treat FILD/FILA as the primary empirical endpoint and the two nonfibrotic outcomes as sparse ancillary checks rather than independent replications.

The empirical analyses are hierarchical. First, repeated held-out splits compare global discrimination and stringent operating points for ZeBRA, genomic-only, and nonlinear combined models. Second, a paired regularized analysis tests incremental genomic value after conditioning on ZeBRA. Third, locus-specific analyses determine whether ZeBRA reconstructs *MUC5B* and whether residual *MUC5B* disease information varies across ZeBRA score space. Fourth, a leakage-free two-threshold rescue analysis tests whether consulting *MUC5B* only near selected ZeBRA boundaries improves held-out sensitivity. Finally, a cross-fitted likelihood-ratio analysis estimates the score-level clinical and residual-genomic evidence channels directly. Computational provenance and exact analysis-output mappings are provided in the Supplementary Methods.

B. Global repeated-split discrimination

Across 30 repeated held-out outer splits, ZeBRA is the strongest predictor for all three operational endpoints. For FILD/FILA, mean AUC is 0.8142 for ZeBRA, 0.5662 for the selected genomic panel, and 0.7988 for the nonlinear combined model. In the paired repeated-split comparison, combined-minus-ZeBRA FILD/FILA AUC is negative in all 30 splits (mean -0.01535). The sparse nonfibrotic endpoints show the same ordering but are not treated as replication.

TABLE I

THEORY–EVIDENCE CORRESPONDENCE. THE EMPIRICAL ANALYSES TEST CONSEQUENCES AND OPERATING BEHAVIOR; THEY DO NOT PROVE THE ASSUMPTIONS GLOBALLY.

Theory component	Empirical analysis	Empirical meaning
Distinct channels and residual genomic evidence	ZeBRA-to- <i>MUC5B</i> pooled and FILD-stratified analyses	Weak global genomic increment is not explained by ZeBRA reconstructing rs35705950.
Finite-stage boundary localization (Theorem 1(ii))	Direct empirical residual-LR estimation, local information curves, and decision-flip localization	The LR analysis estimates a control-side residual-genomic scale at the observed ZeBRA-score stage; it supports finite-stage localization but does not establish uniform tightness over clinical-history length.
Local gain with little global AUC change (Proposition 2)	Repeated-split global comparison, paired incremental analysis, and matched-operational-FPR rescue	Broad global augmentation yields essentially no FILD/FILA AUC gain even though selected boundary decisions improve.
Clinical-tail dominance (Theorem 1(iii); Proposition 1)	Direct clinical-tail LR summaries and stringent held-out operating points	Clinical evidence dominates residual genomic scale over the observed extreme ZeBRA tail. This supports the predicted geometry but does not prove A1-U or mathematical unboundedness in A2.
Scope caveat	Same direct LR analyses	The stochastic bounds are estimated under the control law and for the measured <i>MUC5B</i> residual channel; they are not evidence of a universal genome-wide bound.

TABLE II

PRIMARY COHORT SIZES AND REPEATED-SPLIT AUC. THE TWO NONFIBROTIC ENDPOINTS ARE SPARSE ANCILLARY ANALYSES.

Target	Positives / N	ZeBRA AUC	Genomic AUC	Combined AUC
FILD/FILA	254/12,825	0.8142 ± 0.0089	0.5662 ± 0.0245	0.7988 ± 0.0150
Nonfibrotic ILA	74/12,825	0.8281 ± 0.0162	0.4717 ± 0.0443	0.7112 ± 0.0541
Nonfibrotic ILD	27/12,825	0.8202 ± 0.0257	0.5141 ± 0.0719	0.6145 ± 0.0953

The combined learner was not simply blind to the major genomic signal: in the repeated-split feature-attribution analysis, the ZeBRA score is a top-20 combined-model feature in 30/30 splits, while the rs35705950 GG indicator is top-20 in 28/30 combined-model splits and 27/30 genomic-only splits. Thus the lack of global improvement should not be interpreted as evidence that the fitted genomic models never used *MUC5B*.

C. Paired incremental value conditional on ZeBRA

A paired regularized analysis evaluates ZeBRA-only and ZeBRA-plus-genomics models on the same held-out subjects in 30 repeated splits. For FILD/FILA, the mean paired AUC difference is only $+0.00043$, the median is 0, and the split-wise 2.5th–97.5th percentile range is $[-0.00507, 0.00408]$. Average precision decreases, while Brier score and log loss worsen. These are empirical split quantiles, not confidence limits from independent samples.

D. ZeBRA does not reconstruct *MUC5B*

Among 12,766 subjects with called rs35705950 genotype, ZeBRA predicts T-carrier status at essentially chance level: pooled AUC 0.5088, explicitly adjudicated FILD/FILA-negative AUC 0.4847, and FILD/FILA-positive AUC 0.4817. Carrier prevalence is nevertheless 39.3% in FILD/FILA-positive subjects versus 19.2% among explicit negatives. The pooled top 1% ZeBRA tail shows 1.47-fold carrier enrichment, but the corresponding enrichment is absent within disease strata. This pattern is consistent with shared association to disease status rather than recovery of genotype from clinical history.

E. Residual genomic information is localized in ZeBRA score space

The absence of global incremental AUC does not imply that the genomic channel is conditionally uninformative everywhere. We therefore examined both clinically interpretable control-FPR bands and overlapping moving windows along the ZeBRA percentile axis. Here “conditional LRT” denotes the nested-model likelihood-ratio (deviance) statistic for adding one predictor after the other within a local subset. It is evidence for conditional predictive contribution, not an estimate of the individual-level likelihood-ratio factor $\Lambda_{G|C}$ in Theorem 1. These analyses therefore test a qualitative implication of Theorem 1(ii): residual genomic predictive evidence should be concentrated in restricted clinical-evidence regimes rather than distributed uniformly. They do not estimate the deterministic special-case interval $[\lambda/M, \lambda/m]$ in Corollary 1, and the overlapping windows are descriptive localization analyses rather than independent confirmatory tests.

The FPR-defined analysis makes the directional asymmetry explicit. In every displayed band, the likelihood-ratio statistic for adding *MUC5B* after ZeBRA ($G | Z$) exceeds the converse statistic for adding ZeBRA after *MUC5B* ($Z | G$). The difference is especially large in the 5–10% band, where $G | Z = 16.55$ versus $Z | G = 6.84$, and in the 20–40% band, where $G | Z = 13.69$ versus $Z | G = 0.064$ (Table III). We treat these local statistics as descriptive localization measures rather than confirmatory tests because multiple bands and overlapping score regions are examined in the same cohort; nominal local p-values are therefore not used as primary inferential evidence in the manuscript.

The moving-window analysis in Fig. 3b shows the regime

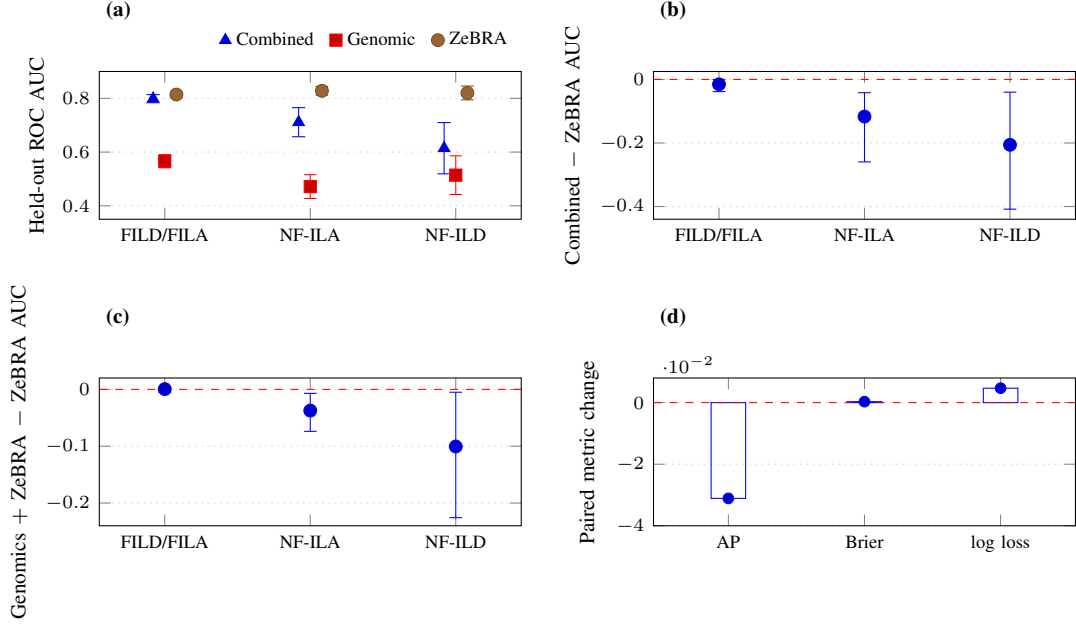


Fig. 1. **Global discrimination and incremental genomic value.** (a) Mean held-out AUC with one standard deviation across repeated outer splits; small horizontal offsets separate the three model estimates at each endpoint. (b) Paired AUC difference between the nonlinear combined model and ZeBRA for each endpoint; negative values favor ZeBRA. (c) Paired AUC change after adding the genomic panel to a regularized ZeBRA model evaluated on the same held-out subjects. (d) Mean paired changes in average precision (AP), Brier score, and log loss for the primary FILD/FILA endpoint. For AP, negative values indicate deterioration; for Brier score and log loss, positive values indicate deterioration. The FILD/FILA endpoint is primary; the two nonfibrotic endpoints are sparse ancillary comparisons.

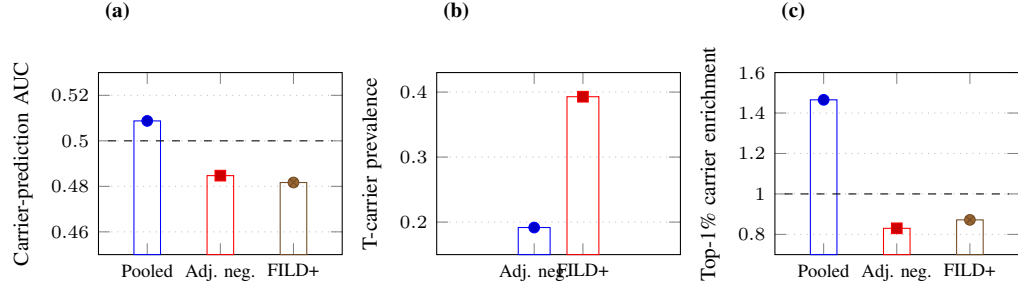


Fig. 2. **ZeBRA and *MUC5B* represent distinct predictive channels.** (a) ZeBRA predicts T-carrier status at approximately chance level in the pooled sample and within disease strata. (b) T-carrier prevalence is substantially higher among FILD/FILA-positive subjects than among explicitly adjudicated-negative subjects. (c) The apparent carrier enrichment in the extreme pooled ZeBRA tail disappears after stratification by disease status; the dashed line denotes no enrichment. Together, these results argue against implicit genotype reconstruction as the explanation for ZeBRA's predictive performance.

transition more directly. Across much of the intermediate-to-high ZeBRA axis, residual *MUC5B* information substantially exceeds residual ZeBRA information. For example, at the 70th-percentile window, $G | Z = 10.93$ while $Z | G \approx 0.002$; at the 90th percentile the corresponding values are 19.78 and 1.23; and at 92.5 they are 20.37 and 0.41. The ordering reverses only in the extreme clinical-risk tail: at the 95th-percentile window, $Z | G = 58.76$ versus $G | Z = 22.86$, and at 97.5 the values are 68.12 versus 15.61. Thus the observed data exhibit a local information crossover: genomics remains conditionally useful through a substantial boundary-adjacent region, whereas the clinical-history channel becomes dominant at the most extreme ZeBRA values. This is qualitative correspondence to the theorem's geometry, not an empirical proof of A1-U or A2.

The same pattern is visible without likelihood-ratio termi-

nology. Within the moving windows, observed FILD/FILA prevalence is consistently higher among *MUC5B* T-carriers than non-carriers (Fig. 3c). At the 70th-percentile window, prevalence is 4.28% in carriers versus 0.98% in non-carriers; at 90 it is 8.27% versus 2.05%; at 95 it is 18.45% versus 7.75%; and at 97.5 it is 20.48% versus 9.89%. Local adjusted odds ratios provide the corresponding effect-size view (Fig. 3d): the *MUC5B* carrier OR remains several-fold across much of the risk axis, whereas the ZeBRA OR per local SD rises sharply only in the extreme tail, reaching 2.95 at the 95th percentile and 4.07 at 97.5. These complementary representations make the central empirical claim precise: the genomic signal is neither globally redundant nor uniformly useful; it is locally informative until sufficiently strong clinical evidence becomes dominant. The within-band ZeBRA AUC of 0.367 in the 5–10% control-FPR band is therefore interpreted as a genuine local reversal inside

a narrow score slice rather than as a transcription error or contradiction of ZeBRA’s global performance.

F. Direct empirical likelihood-ratio analysis supports the stochastic theorem

The earlier moving-window statistics describe conditional predictive contribution but are not themselves estimates of the individual-level likelihood-ratio factor in the theorem. We therefore instantiate the likelihood-ratio factorization directly at the observed ZeBRA-score level. Let Z denote the ZeBRA score and $G \in \{0, 1\}$ indicate *MUC5B* T-carrier status. Then, without any sufficiency assumption,

$$\Lambda_{Z,G}(z, g) = \Lambda_Z(z) \Lambda_{G|Z}(g; z), \quad (44)$$

where

$$\Lambda_Z(z) = \frac{p(z | Y = 1)}{p(z | Y = 0)}, \quad (45)$$

$$\Lambda_{G|Z}(g; z) = \frac{p(g | Y = 1, Z = z)}{p(g | Y = 0, Z = z)}. \quad (46)$$

For carrier status, defining $q_y(z) = \mathbb{P}(G = 1 | Y = y, Z = z)$ gives

$$\Lambda_{G|Z}(1; z) = \frac{q_1(z)}{q_0(z)}, \quad \Lambda_{G|Z}(0; z) = \frac{1 - q_1(z)}{1 - q_0(z)}. \quad (47)$$

These quantities are estimated by five repeats of fivefold cross-fitting with cubic spline logistic models. The primary specification uses five spline knots and logistic regularization $C = 1$; a prespecified sensitivity grid uses four, five, or six knots crossed with $C \in \{0.3, 1, 3\}$. For each control subject, the statewise residual-genomic scale is

$$B(Z) = \max_{g \in \{0, 1\}} \left| \log \hat{\Lambda}_{G|Z}(g; Z) \right|, \quad (48)$$

and the empirical stochastic bound is the $(1 - \delta)$ control quantile $\hat{b}_\delta = Q_{1-\delta}[B(Z) | Y = 0]$. This is a score-level empirical analogue of the finite-stage scale $b_{\delta,n}$ in Theorem 1(ii). Here $Z = f(C_n)$ is a one-dimensional summary of the observed clinical history. The score-level factorization $\Lambda_{Z,G} = \Lambda_Z \Lambda_{G|Z}$ is exact and requires no sufficiency assumption, so the analysis is a valid theorem instance for the observed clinical channel Z . It does not establish A1-U, a population-wide bound for the complete clinical history C_n , or a bound for arbitrary genomic features beyond the measured *MUC5B* channel. If Z retained all disease-relevant information in C_n needed for the residual genomic likelihood ratio—equivalently, if conditioning on Z and on C_n induced the same residual genomic law—then $\Lambda_{G|Z}$ would coincide with $\Lambda_{G|C_n}$. We neither assume nor establish that stronger sufficiency property.

In the primary 5-knot/ $C=1$ specification, the control-side high-probability residual-genomic bounds are $b_{0.05} = 0.8699$, $b_{0.01} = 0.9099$, and $b_{0.005} = 0.9241$ on the log-LR scale. Across all nine specifications, the corresponding ranges are 0.8441–0.9083, 0.8635–0.9617, and 0.8697–1.6100. Thus the 95% and 99% bounds are stable, whereas the 99.5% bound becomes unstable only in the most flexible 6-knot fits. This pattern supports a high-probability finite-stage formulation rather than a literal deterministic bound. It should not be

interpreted as evidence that one common residual-genomic bound holds uniformly over all clinical-history lengths.

The clinical-tail comparison is similarly direct. In the primary specification, median clinical log-LR among subjects above the 95th, 97.5th, and 99th ZeBRA percentiles is 1.5175, 2.7325, and 3.8345, while the corresponding 95th percentiles of the estimated statewise residual genomic scale are only 0.9213, 0.7758, and 0.7384. The fraction of subjects whose clinical log-LR exceeds that genomic scale is 0.9045, 0.9938, and 1.0000, respectively. Across all nine specifications, the analogous fraction equals 1.0 at the 99th percentile cutoff in every case.

Finally, decision flips induced by genomics remain localized near the clinical boundary. Across the nine specifications, the median distance of flipped points from the clinical boundary is 0.1345–0.3039 log-LR units at requested FPR 0.5% and 0.2292–0.4991 at requested FPR 1%; the corresponding numbers of flipped controls are only 2–7 and 21–69. Together these direct LR-based results provide the main empirical support for finite-stage boundary localization and for clinical dominance over the observed extreme score range. They do not establish A1-U or prove the unbounded-tail assumption A2.

G. Boundary-targeted *MUC5B* use: held-out gain at matched false-positive rate

The two-threshold rescue rule is evaluated leakage-free: upper and lower ZeBRA thresholds are derived from training negatives, the *MUC5B* rescue is applied only to untouched outer-test subjects, and a ZeBRA-only threshold is separately selected on training data to match the rescue rule’s training-set FPR before test evaluation. Across all six evaluated threshold pairs, the *mean* held-out sensitivity gain over this fair matched-ZeBRA comparator is positive. For the 0.5%/10% rule, mean sensitivity increases by 0.0446 while mean operational FPR differs from matched ZeBRA by only -2.53×10^{-4} .

We additionally resolved the stricter adjudicated-negative comparison that was previously a limitation. Each original outer split was reconstructed from its saved split seed, and the already training-derived matched-ZeBRA threshold was applied unchanged to the explicitly adjudicated-negative subjects in the untouched test set; no threshold was fitted within that subset. For the 0.5%/10% rule, mean adjudicated-negative FPR is 0.1292 for the original stringent ZeBRA rule, 0.1958 for rescue, and 0.2144 for the fair matched-ZeBRA comparator. Thus rescue increases FPR relative to the original more-stringent baseline, but is 0.0185 lower than matched ZeBRA on average. Rescue has lower mean adjudicated-negative FPR than matched ZeBRA in five of the six evaluated rules; the remaining 1%/2% rule is essentially equal (+0.00065). For the 0.5%/10% rule, rescue has lower adjudicated-negative FPR than matched ZeBRA in 81% of outer splits. Because each split contains only about 216 explicitly adjudicated negatives on average, the split-wise 2.5th–97.5th percentile range for the rescue-minus-matched difference remains broad (-0.0561 to $+0.0194$ for the 0.5%/10% rule). We therefore interpret this analysis as resolving the direction of the earlier diagnostic concern rather than as definitive evidence of a lower adjudicated-negative FPR.

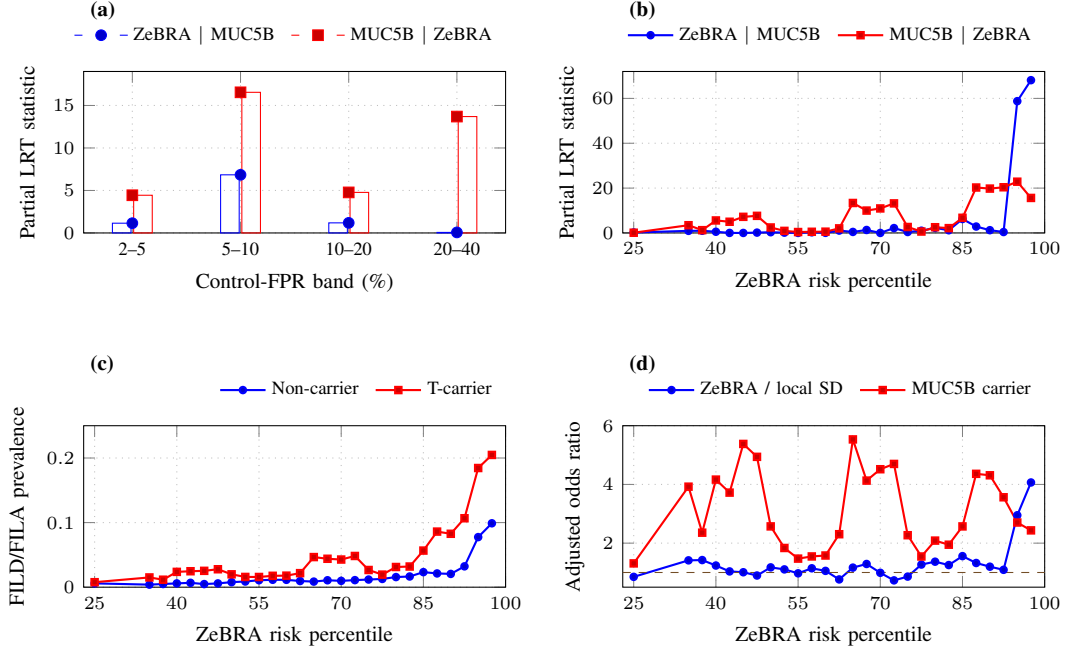


Fig. 3. **Localized complementarity across the ZeBRA risk axis.** (a) Directional nested-model likelihood-ratio statistics in clinically relevant control-FPR bands. (b) Overlapping moving windows show substantial residual *MUC5B* information through intermediate and high ZeBRA percentiles, followed by a sharp rise of the residual ZeBRA contribution in the extreme clinical-risk tail. (c) Within the same score axis, observed FILD/FILA prevalence is consistently higher among *MUC5B* T-carriers than non-carriers. (d) Local adjusted odds ratios provide an effect-size view of the same transition. Moving windows overlap and are used for descriptive localization rather than as independent hypothesis tests; no nominal significance threshold is shown.

TABLE III

LOCAL FILD/FILA INFORMATION IN ZEBRA-DEFINED CONTROL-FPR BANDS. BOTH DIRECTIONAL CONDITIONAL LRTs ARE SHOWN SO THE ASYMMETRY BETWEEN CHANNELS IS EXPLICIT. THESE LOCAL SUMMARIES ARE DESCRIPTIVE RATHER THAN MULTIPLICITY-ADJUSTED CONFIRMATORY TESTS.

Band	<i>N</i>	Cases	LRT <i>Z</i> <i>G</i>	LRT <i>G</i> <i>Z</i>	<i>G</i> – <i>Z</i> partial	AUC ZeBRA	AUC <i>MUC5B</i>
2–5%	401	26	1.15	4.44	3.29	0.577	0.595
5–10%	652	26	6.84	16.55	9.71	0.367	0.677
10–20%	1,281	30	1.19	4.77	3.58	0.558	0.586
20–40%	2,540	37	0.06	13.69	13.63	0.516	0.635

TABLE IV

SELECTED MOVING-WINDOW VALUES ILLUSTRATING THE INFORMATION CROSSOVER. WINDOWS ARE CENTERED ON THE INDICATED ZEBRA PERCENTILE AND OVERLAP; VALUES ARE DESCRIPTIVE LOCALIZATION SUMMARIES.

ZeBRA percentile	LRT <i>Z</i> <i>G</i>	LRT <i>G</i> <i>Z</i>	OR ZeBRA/local SD	OR <i>MUC5B</i>	Prev. carrier	Prev. non-carrier
65.0	0.45	13.36	1.17	5.53	0.0466	0.00865
70.0	0.002	10.93	0.99	4.52	0.0428	0.00980
90.0	1.23	19.78	1.20	4.31	0.0827	0.0205
92.5	0.41	20.37	1.09	3.56	0.1067	0.0323
95.0	58.76	22.86	2.95	2.70	0.1845	0.0775
97.5	68.12	15.61	4.07	2.43	0.2048	0.0989

TABLE V

BOUNDARY-TARGETED *MUC5B* RESCUE COMPARED WITH A FAIR ZEBRA-ONLY RULE MATCHED ON TRAINING-SET OPERATIONAL FPR. SENSITIVITY AND FPR DIFFERENCES ARE RESCUE MINUS MATCHED ZEBRA. THE ADJUDICATED-NEGATIVE COMPARISON USES THE SAME UNTOUCHED OUTER-TEST SUBJECTS AND THE ALREADY TRAINING-DERIVED MATCHED THRESHOLD. VALUES ARE MEANS ACROSS 100 REPEATED OUTER SPLITS; THE FINAL COLUMN GIVES THE FRACTION OF SPLITS IN WHICH RESCUE PRODUCED A LOWER ADJUDICATED-NEGATIVE FPR THAN MATCHED ZEBRA.

Upper/lower target FPR	Δ sensitivity (pp)	Δ operational FPR (pp)	Δ adjudicated FPR (pp)	Rescue lower (%)
0.5% / 2%	+1.81	+0.0012	-0.37	51
0.5% / 5%	+3.66	-0.0040	-0.60	61
0.5% / 10%	+4.46	-0.0253	-1.85	81
1% / 2%	+0.96	+0.0007	+0.06	31
1% / 5%	+2.25	-0.0184	-0.46	55
1% / 10%	+3.35	-0.0178	-1.92	78

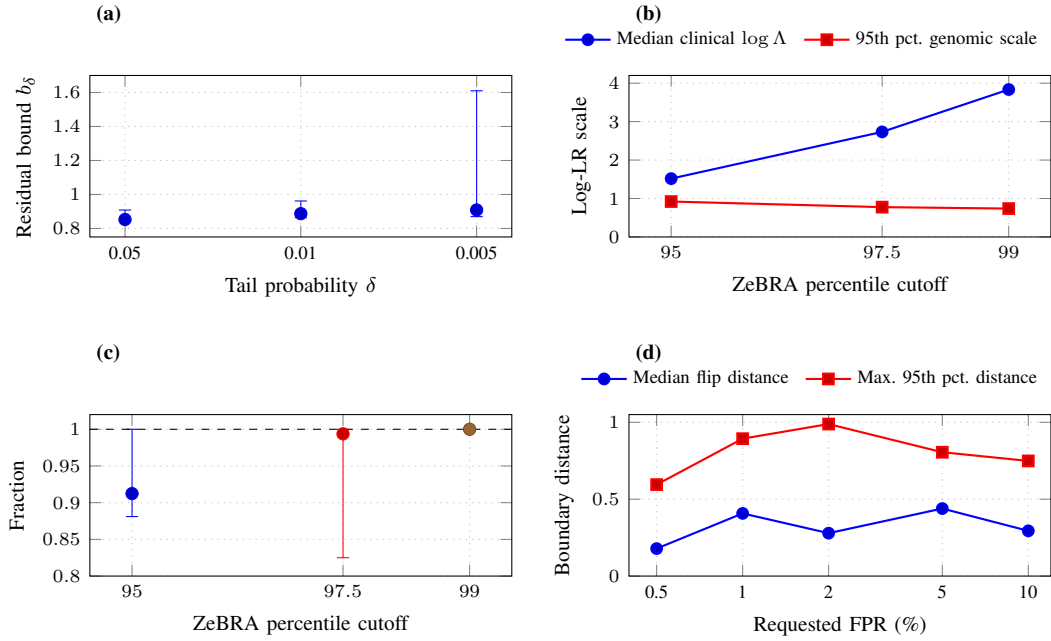


Fig. 4. **Direct empirical likelihood-ratio support for stochastic boundary localization.** (a) Median high-probability residual-genomic bounds across nine prespecified estimator specifications, with error bars spanning the minimum to maximum. The 95% and 99% bounds are stable, whereas the 99.5% tail is more estimator-sensitive. (b) In the primary estimator, median clinical log-likelihood evidence rises sharply across the upper ZeBRA tail while the 95th percentile of the residual genomic scale remains substantially smaller. (c) The fraction of subjects for whom clinical evidence exceeds the estimated residual genomic scale approaches one in the extreme tail; points show the median across specifications and error bars span the full sensitivity range. (d) Genomic-induced decision changes remain concentrated near the clinical likelihood-ratio boundary across stringent operating points.

H. Stringent operating points provide direct clinical-tail correspondence

The stringent held-out operating-point analysis provides a direct empirical comparison to the clinical-tail portion of the theory that is more informative than global AUC alone. At requested FPR 0.5%, mean held-out LR+ is approximately 57.9 for ZeBRA, 4.28 for the genomic-only model, and 39.1 for the nonlinear combined model. At requested FPR 1%, the corresponding means are 29.8, 3.62, and 26.7. Thus the evaluated clinical-history score dominates the evaluated genomic-only model in the stringent tail. Because raw ZeBRA is not calibrated onto a likelihood-ratio scale in this cohort, these LR+ values characterize the induced held-out decision rules rather than directly identifying a threshold K on Λ_C . They therefore provide the appropriate operating-point correspondence to the theorem without proving either an unbounded clinical tail or a universal genomic bound M_G .

The supporting hybrid ROC analysis is included only as a geometric complementarity diagnostic. The empirical hybrid AUC is 0.8095 versus 0.8142 for ZeBRA; the convex hull of the union of the observed ZeBRA and hybrid ROC points has descriptive AUC 0.8260. This hull is not a validated classifier and should not be interpreted as held-out model performance or as a universal theoretical maximum. It is the best randomized operating envelope attainable from the evaluated ROC point set and illustrates where the two score families offer complementary operating points.

I. Calibration diagnostics delimit the present theory–data mapping

The raw ZeBRA score is not calibrated as an absolute FILD/FILA probability in this cohort. Across repeated held-out splits, the score has mean calibration slope approximately 0.18 and expected calibration error approximately 0.57; its mean value is about 0.59 while operational FILD/FILA prevalence is about 0.020. We therefore treat ZeBRA empirically as an ordinal clinical-history score in the current paper. Corollary 2, which maps a calibrated posterior probability to a likelihood ratio, therefore remains a theoretical result and is not numerically tested with the raw ZeBRA score. The empirical theory–data connection instead uses the directly estimated score-level likelihood ratio Λ_{Z_i} .

Calibration summary for FILD/FILA		
Model	Calibration slope	ECE
ZeBRA	0.180	0.570
Genomic	0.241	0.115
Combined	1.140	0.121

VIII. SCOPE AND LIMITATIONS

The theory is a statement about information channels, not about a particular machine-learning architecture. It does not require independence between genomics and clinical history; dependence is retained explicitly through $\Lambda_{G|C}$. The marginal genomic power result also propagates through deterministic or class-independent randomized post-processing of the measured genomic state, as formalized in Corollary 3.

Several qualifications are essential. First, the direct empirical likelihood-ratio analysis estimates a finite-stage control-side

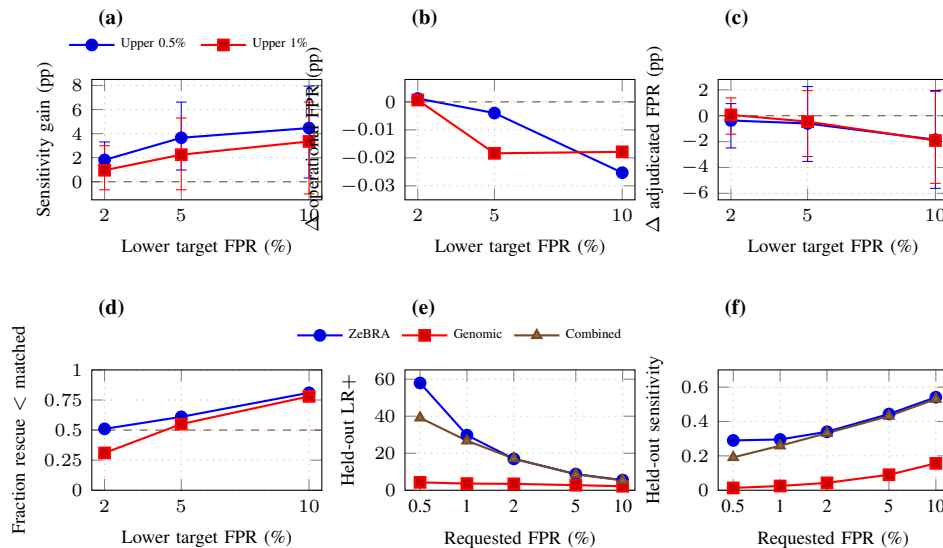


Fig. 5. **Decision-level consequences of localized genomic utility and clinical-tail dominance.** (a) Held-out sensitivity gain for boundary-targeted *MUC5B* rescue relative to a ZeBRA-only threshold matched on training-set false-positive rate. (b) The corresponding difference in realized operational false-positive rate remains near zero. (c) Among explicitly adjudicated negatives, rescue-minus-matched-ZeBRA false-positive-rate differences are negative on average for five of the six evaluated rules; error bars show split-wise 2.5th–97.5th percentile ranges. (d) Fraction of outer splits in which rescue has lower adjudicated-negative FPR than its fair matched-ZeBRA comparator; the dashed line marks 0.5. Circles denote an upper target FPR of 0.5% and squares 1% in panels (a)–(d). (e) At stringent requested false-positive rates, ZeBRA achieves substantially larger held-out LR+ than the genomic-only model. (f) Corresponding held-out sensitivity.

residual-genomic scale for the measured *MUC5B* channel. It does not establish the optional uniform-tightness condition A1-U across clinical-history lengths, does not establish the same bound under the case law, and does not imply a universal genome-wide bound. Second, the deterministic bounded-residual corollary remains a useful idealization but is a stronger special case. Third, A2 is an explicit assumption about the existence of arbitrarily strong clinical-evidence tails; sequential factorization makes accumulation possible but does not prove unboundedness, and the Colorado data demonstrate only a strong observed tail over a finite range. Fourth, Theorem 1(ii) concerns threshold reversals, not all posterior-risk changes. Fifth, the current genomic comparator is a selected panel and the direct residual-LR analysis focuses on *MUC5B*; neither should be interpreted as an empirical bound on genome-wide predictors, external polygenic risk scores, or unmeasured genomic information. Sixth, the raw ZeBRA score is not calibrated as an absolute posterior probability in this cohort, so Corollary 2 is not numerically applied; the empirical connection instead uses direct score-level LR estimation. Seventh, the overlapping local windows and FPR bands are descriptive localization analyses, not confirmatory independent tests; nominal local p-values are not used as primary evidence. Eighth, the operational FILD/FILA endpoint includes missing adjudication values in the target-zero comparison group. We therefore evaluated the fair matched-ZeBRA comparator separately within explicitly adjudicated negatives. Rescue has similar or lower mean adjudicated-negative FPR across the six rules, including a 1.85-percentage-point lower mean FPR for the 0.5%/10% rule, but the subset contains only about 216 negatives per split and the split-wise percentile intervals cross zero; this result should therefore be interpreted as a resolved control diagnostic rather than definitive superiority in

the adjudicated population. Finally, the nonfibrotic endpoints contain only 74 and 27 positives and are treated solely as sparse ancillary checks, while all rescue and LR analyses remain internal and require prospective or external validation.

A stronger future validation should therefore calibrate ZeBRA explicitly onto a likelihood-ratio or posterior-probability scale, estimate uncertainty for the direct LR quantities, evaluate prespecified fixed-specificity thresholds prospectively, enlarge or externally replicate the explicitly adjudicated-negative comparison, and replicate boundary-localized genomic rescue in an external cohort.

IX. CONCLUSION

The central result is not that clinical history “contains” genomics, nor that genomics is globally useless. The two modalities are distinct evidence channels. Bayes’ rule factorizes their joint evidence into a clinical likelihood ratio and a residual genomic likelihood ratio conditional on clinical state. The key theoretical refinement is that finite-stage boundary localization is stated directly in terms of the high-probability residual-genomic scale at the clinical stage being analyzed; it does not require a bound uniform over all history lengths. Uniform tightness is introduced only as an optional stronger assumption for sequential clinical-tail dominance. Under the deterministic special case, the exact boundary band is recovered; under a marginal genomic likelihood-ratio bound, sufficiently extreme clinical evidence dominates any genomic-only rule whenever the corresponding clinical-tail condition holds.

The current Colorado results show direct empirical correspondence with this geometry. ZeBRA dominates global discrimination; broad genomic augmentation provides essentially zero incremental FILD/FILA AUC after conditioning on ZeBRA; ZeBRA does not reconstruct *MUC5B*; residual *MUC5B* disease

information is heterogeneous across selected ZeBRA operating bands; a boundary-targeted rescue rule improves mean sensitivity and LR+ at closely matched operational FPR and does not show excess mean FPR relative to the fair matched-ZeBRA comparator within explicitly adjudicated negatives; and the direct LR analysis estimates stable 95% and 99% control-side residual-genomic scales, demonstrates that clinical log-LR clearly exceeds residual genomic scale in the extreme observed ZeBRA tail, and shows that genomic-induced decision flips remain close to the clinical boundary. The resulting picture is one of *regime-dependent complementarity*: genomics can matter locally and operationally even when its global incremental discrimination is minimal.

Because the direct LR analysis remains internal to the current cohort and focuses on the measured MUC5B residual channel, it should be understood as empirical support for finite-stage boundary geometry rather than as proof of A1-U, A2, or universal genomic boundedness. Even so, the revised theory and data now align more tightly: the assumptions are explicitly separated by strength, and the central claims are supported by figures and tables that display the relevant likelihood-ratio quantities without overextending the empirical scope.

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